

Project Title: Design RDBMS for Aerox

Objective:

1. Design RDBMS Schema for Aerox and Understand the trend analysis
 2. Understand operational performance of Aerox
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CREATE DATABASE:

```
create database if not exists Aerox;
```

USE DATABASE:

```
use Aerox;
```

CREATE TABLE PASSENGERS:

```
create table passengers (  
  passenger_id int primary key unique not null,  
  passenger_fullname varchar(30) not null,  
  passenger_mobilenno bigint not null,  
  passenger_email varchar (30) not null,  
  passenger_passport_no varchar (20) not null);
```

CREATE TABLE BOOKINGS:

```
create table bookings(  
  booking_id int primary key unique not null,  
  booking_date datetime not null,  
  booking_class enum ("First Class", "Economy", "Business", "Cargo") not  
  null,  
  seat_available int not null,  
  seat_no int not null,  
  booking_status enum ("Confirm", "Pendiing", "Canceled") not null,  
  passenger_id int not null,  
  foreign key (passenger_id) references passengers (passenger_id));
```

CREATE TABLE ROUTES:

```
create table routes(  
  route_id varchar(20) primary key unique not null,  
  route_source varchar (20) not null,  
  route_destination varchar (20) not null,  
  route_start_time datetime not null,  
  route_end_time datetime not null,  
  route_status enum ("Active", "Inactive", "Scheduled", "Delayed",  
  "Completed") not null,  
  route_distance_km int not null,  
  passenger_id int not null,  
  foreign key (passenger_id) references passengers (passenger_id),  
  booking_id int not null,  
  foreign key (booking_id) references bookings (booking_id));
```

CREATE TABLE AIRFLIGHTS:

```

create table airflights(
airflight_id varchar(20) primary key unique not null,
airflight_model varchar(20) not null,
airflight_capacity int not null,
airflight_status enum ("Active", "Maintenance", "Retired") not null,
passenger_id int not null,
foreign key (passenger_id) references passengers (passenger_id),
booking_id int not null,
foreign key (booking_id) references bookings (booking_id),
route_id varchar(20) not null,
foreign key (route_id) references routes (route_id));

# CREATE TABLE AIRPORTS:

create table airports(
airport_code varchar (20) primary key unique not null,
airport_name varchar (20) not null,
airport_city varchar(20) not null,
airport_country varchar(20) not null,
airport_type enum ("International", "Domestic") not null,
booking_id int not null,
foreign key (booking_id) references bookings (booking_id),
airflight_id varchar (20) not null,
foreign key (airflight_id) references airflights (airflight_id));

# CREATE TABLE CREW_INFO:

create table crew_info(
crew_id int primary key unique not null,
crew_fullname varchar(50) not null,
crew_mobilno bigint not null,
crew_email varchar(50),
crew_designation varchar (20) not null,
route_id varchar(20) not null,
foreign key (route_id) references routes (route_id),
airflight_id varchar (20) not null,
foreign key (airflight_id) references airflights (airflight_id),
airport_code varchar (20) not null,
foreign key (airport_code) references airports (airport_code));

# CREATE TABLE SERVICES:

create table services(
ticket_id varchar (20) primary key unique not null,
maintenance_status enum ("scheduled", "In-Progress", "Complete", "Pending")
not null,
security_status enum ("Cleared", "Under check", "Restricted") default
"Under check",
service_type enum ("Inflight", "Ground", "Premium") not null,
service_name enum ("Inflight meal", "WIFI", "Extra baggage",
"Entertainment")not null,
passenger_id int not null,
foreign key (passenger_id) references passengers (passenger_id),
booking_id int not null,
foreign key (booking_id) references bookings (booking_id),
route_id varchar(20) not null,
foreign key (route_id) references routes (route_id),

```

```

airflight_id varchar (20) not null,
foreign key (airflight_id) references airflights (airflight_id));

# CREATE TABLE TRANSACTIONS:

create table transactions(
transaction_id bigint primary key unique not null,
transaction_amount decimal(10,2) not null,
transaction_method enum ("UPI", "Net Banking", "Wallet", "Cards") not null,
transaction_date datetime not null,
transaction_status enum ("Complete", "Pending", "Failed", "Canceled",
"Refunded", "Processing") not null,
booking_id int not null,
foreign key (booking_id) references bookings (booking_id),
ticket_id varchar(20) not null,
foreign key (ticket_id) references services (ticket_id),
passenger_id int not null,
foreign key (passenger_id) references passengers (passenger_id));

# CREATE TABLE REVENUE:

create table revenue(
revenue_id int primary key unique not null,
revenue_amount decimal(10,2) not null,
ticket_id varchar (20) not null,
foreign key (ticket_id) references services (ticket_id),
booking_id int not null,
foreign key (booking_id) references bookings (booking_id),
transaction_id bigint not null,
foreign key (transaction_id) references transactions (transaction_id));

# CREATE TABLE REVIEWS:

create table reviews(
passenger_feedback varchar(200),
passenger_rating int check (passenger_rating between 1 and 5),
crew_feedback varchar(200),
crew_rating int check (crew_rating between 1 and 5),
booking_feedback varchar (200),
passenger_id int not null,
foreign key (passenger_id) references passengers (passenger_id),
booking_id int not null,
foreign key (booking_id) references bookings (booking_id),
crew_id int not null,
foreign key (crew_id) references crew_info (crew_id));

```